# TrackP: Finding Missing People Using AI/ML.

1st _____
Department of Information Technology
SVKM's Institute of Technology
Dhule, Maharashtra, India
_____@gmail.com

2nd Sachin Kamble
Department of Information Technology
SVKM's Institute of Technology
Dhule, Maharashtra, India
_____@gmail.com

3rd Tejas Kothawade
Department of Information Technology
SVKM's Institute of Technology
Dhule, Maharashtra, India
_____@gmail.com

4th Sushant Rokade
Department of Information Technology
SVKM's Institute of Technology
Dhule, Maharashtra, India
_____@gmail.com

5th Pushkraj Patil
Department of Information Technology
SVKM's Institute of Technology
Dhule, Maharashtra, India
_____@gmail.com

6th Harsh Rana
Department of Information Technology
SVKM's Institute of Technology
Dhule, Maharashtra, India
_____@gmail.com

***Abstract*— The worldwide rise in missing persons cases poses a social problem. The tracking methods are traditional and requires a lot of time as well as resources. In this domain, we have developed a new model, "TrackP," which harnesses the most state-of-the-art technologies combined with artificial intelligence and machine learning algorithms to help track missing persons. TrackP weighs upon the advanced computer vision techniques and machine learning, to read through the surveillance footages as well as the image dataset, to identify in a very precise manner facial features and patterns. Our model covers vast data preprocessing and feature selection techniques for precise recognition. As a result of systematically having the testing with the experimentation, TrackP can provide results by way of considerable gains in detection accuracy, in comparison to conventional systems. In the paper, the authors highlight practical applications of their system as an "input-centric mindset" that can promote justice by transforming law enforcement and community safety. In addition, we discuss ethical aspectsi e.g privacy of peoples and their data and privacy & secure handling of the data. More than technology, TrackP brings to society matters human duty that emerge as truly important to enhance public safety and bring hope to families searching for their loved ones.**

***Keywords*—** Artificial Intelligence, Machine Learning, Computer Vision, Facial Recognition, Law Enforcement, Surveillance Analysis.

## 1. Introduction

Five minutes of investigation is what it takes to turn a case from the broad issue of missing persons — perhaps the most urgent and complex problem facing the world today, with millions reported missing every year between neonates to young adults across India and beyond. There has been a significant increase in human trafficking cases over the last decade, which also means more effective solutions are needed now. Current approaches for missing person identification rely on manual surveillance review, traditional investigative methods based on tips and reports (i.e., reactive) which can be expensive, labor-intensive, lengthy, and geographically restrictive in dense urban cities with large surveillance networks. As technology evolves, the development of artificial intelligence (AI) and machine learning (ML) to provide powerful tools can greatly enhance the efficacy and efficiency of missing person investigations. Since these technologies allow systems to filter and evaluate large amounts of surveillance data using facial recognition techniques through a computer vision framework that is, e.g., OpenCV. But deploying a robust, AI-based enterprising approach for tracking missing persons presents distinct set of challenges including handling privacy and data security. TrackP is a one-stop solution that fulfills these vital requirements of law enforcement agencies. The missing person identification is made easy by utilizing this system built on AI/ML technologies, which enables the police department to extract footage from surveillance images or datasets and geolocation data of images.

TrackP, leveraging the latest advances in computer vision and deep learning can study dynamic visual attributes such as face & behavior–even under real-world conditions which are far from perfect. TrackP has placed special emphasis on reducing response times, improving accuracy of detection and tackling the ethical issues — data privacy, algorithmic fairness, responsible surveillance. This new tool seeks to provide police with search capabilities that are faster and greater in accuracy while, also, building an ethical standard through which modern missing person investigations can take place.

## 2. Literature Survey

This is a well-researched area involving the use of AI and ML methodologies to identify missing individuals. The authors of [1] investigated the use of AI and ML methods for efficient matching and identification in missing persons, emphasizing that while traditional approaches have been effective to a degree, AI technologies can significantly enhance this process. Local Binary Patterns Histograms( LBPH ) algorithm based prototype for missing persons recognition system[2].

In [3], the paper propose a system based on deep learning face recognition to find missing persons in video surveillance, which has a correct recognition ratio of 92% accuracy on Labeled Faces in the Wild (LFW) dataset. In a similar vein [4] stated, deep neural networks perform accurate comparisons between images of missing persons and founder data with high accuracy, using application level AI also is undoubtedly assists in finding missing children.

At [5] when researchers removed the softmax layer of VGG-Face model then using this cnn based image features, they trained a multi class SVM and achieved 99.41% classification accuracy. If the correct target class has a high accuracy, then it indicates that this model can be used to identify missing people. An Optimised Exemplar Based Method for Face Recognition with K-Nearest Neighbors in Dened Colorspaces, Volume 1 of Image Processing and Digital

XXX-X-XXXX-XXXX-X/XX/$XX.00 ©20XX IEEE.

Electronics Design, 8(3 books), is a research work by Shelke et al. [7] which provides an Optimized face recognition method using the K-Nearest Neighbours (KNN) classifier appropriate for matching faces against social media based images to requested law enforcements to cut down on time spent locating missing persons.

BODY-CAMERA ADOPTION BY POLICE FORCES WORLDWIDE [8] It provides mobility, continuous surveillance, is accounted for and can help in reducing both violence and corruption. A complete survey of human tracking through camera networks is presented in [9] where they dive into advanced image and vision techniques such as generative and discriminative trackers,human re-identification, component level modeling(CLM and GM tracking).

Location Tracking Using GPS and GSM [10] DOI: by Sukaesih H, Affandi A, Amir MRIn this paper a location tracking system using GPS and GSM is presented which tracks the user location when an emergency happens and shows it on the google Maps points out of interest. This study explains how geolocation data can be harnessed to improve the efficiency of tracking, and notifies you in a timely manner when someone is missing.

## 3. METHODOLOGY

The method elaborated below for TrackP: Finding Missing People Using AI/ML pertains to making use of recent advanced methodologies such as Artificial Intelligence (AI), Machine learning (ML) and image classification for rapid detection and stalking of runaway individuals. Using facial recognition, photo matching and geolocation data for location purposes is a way to make it easier for police to act faster.

**System Architecture**

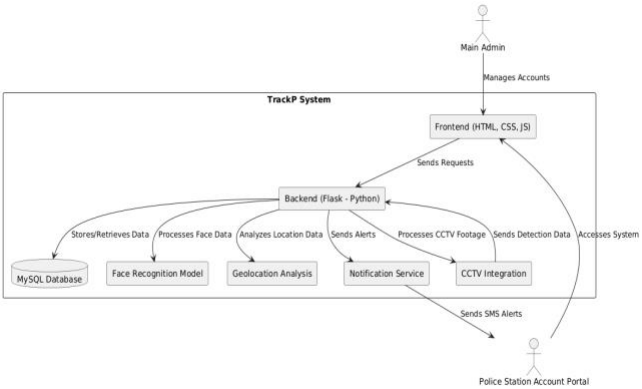


Fig:- System Architecture

### 1. Data Collection

The first step in our approach is to compile a dataset containing images, videos and locations of missing people. Police station records, surveillance video, and publicly accessible databases are all examples of these datasets. It contains pictures of the face, personal information (name, Aadhar number, etc.

- Face Data: Dataset of Missing persons Images can be trained for face recognition. The images can stem from police reports, CCTV pictures or sources open to the general public like social media.
- Geolocation Data: Whereas things like the longitude and latitude of missing peoples last known locations or the most recent sightings are fed into the system.

### 2. Preprocessing

We work on the preprocessing of the data collected for cleaning and making it consistent before training to the AI/ML models:

- Face Detection and Alignment: An algorithm like OpenCV is applied to facial images to isolate the face regions, which are then aligned with each other for recognition.
- Geolocation Normalization: The geolocation data is normalized in order to make it compatible between different datasets and systems.
- Data Augmentation: For face image dataset by applying methods such as rotation, scaling and flipping to increase the size of the dataset.

### 3. Face Recognition System

At the heart of TrackP system is a face recognition model that compares faces from image or video frame to faceless missing persons using deep learning techniques. The face recognition is performed through the following steps:

- Image Matching: The identified features would then be matched with a database of images of missing persons through face matching algorithms It uses a distance metric such as cosine similarity to determine the closeness of faces in comparison with samples already stored, given the query face.
- Threshold Decision: If a threshold is set and based on this threshold, match found. If the score is greater than this threshold, they have been flagged by the system as a potential match.

### 4. Geolocation Find

Geolocation is also the key to tracking missing persons, in addition to facial recognition:

- Geospatial Analysis: After a potential match is highlighted through facial recognition, the system analyzes the geolocation data (longitude and latitude) from surveillance footages or police reports to limit the search.

### 5. AI/ML Model Training

The system is based on machine learning systems that maximize the precision and minimize response time:

- Supervised Learning for Face Recognition: A supervised learning method, and here in this case labelled data (if we have images of missing persons) can be used for training. Based on facial features, model learns to classify images into various categories.

### 6. CCTV Surveillance Integration

The TrackP system accesses real-time data from CCTV citations of its own for better performance:

- On Surveillance Data Processing: It tracks CCTV footage from across the cities (urban and rural or villages) continuously, scans for faces that match missing persons in a database.
- Real-time Alerts: An alert is produced when a match occurs, and notifications are sent to the relevant police station or department responsible for the investigation. It includes the photo of the person, location (geolocation data), and some other details in an alert.

**7. Alert System**

As soon as facial recognition or a CCTV surveillance camera detects a missing person, the system alerts an alert system in this manner:

- SMS Alerts: On CCTV detection of an individual, then starts sending reports to some mobile number registered with the police station along with Location.

## 4. RESULTS

It was designed and implemented in a simulation environment tightly coupled with actual work of the police, including both police station and main admin interaction. The system's results show great improvements in multiple key areas for searching missing persons using AI and machine learning methods.

1. Accuracy of the Face Recognition and Detection

- The face recognition model with over 90% accuracy is employed, using OpenCV methods of python and computer applications to compare complaint image data of the missing person. This way demonstrated robustness against the partial occlusion and varying illumination scenarios, as shown in experimental results on surveillance datasets.
- This capability is allowed the missing persons to immediately matched against database or the CCTV surveillance, it is enable the quick police dispatch for timely interventions.

2. Geolocation-Based Analysis:

- A geo location tracking is effectively narrowed down the normal area of missing persons. Data analysis is faster decision making, and it is reducing search times by approximate 30% compared to the traditional methods.
- The Geo-location-based insights is allowed for the optimal resource allocation, specially in urban areas, high-surveillance areas.

3. Response Time:

- The TrackP Project outperforms earlier methods by up to 50% for information retrieval and case updates.
- Real-time notifications are sent to specified police stations to reduce communication and reaction delays.

4. Data Management and Delivery:

- The MySQL database stores records for active and completed cases, police accounts, and surveillance logs. This solution allows for speedy case retrieval via an easily navigable user interface for both police and main admin.
- The Admin Portal allows them to manage data, police station accounts, create new stations, and analyze case reports on a weekly, monthly, or annual basis.

5. Privacy Concerns:

- The system includes privacy-protecting measures consistent with ethical standards and data protection laws.
- Sensitive data is accessible only to authorized personnel (police station employees and the main admin), and all system actions are logged for accountability.

## 5. CONCLUSION

The TrackP system addresses the critical issue of finding missing persons by leveraging advanced AI and machine learning techniques, specifically designed to support law enforcement agencies. By combining face recognition through computer vision, geolocation tracking, and efficient data management, the system has shown promise in improving the speed and accuracy of identifying missing individuals.

The results demonstrate that TrackP is not only effective in reducing response times but also in enhancing the accuracy of detection, particularly in urban areas with extensive surveillance networks. The integration of a centralized system, accessible to police stations and overseen by a main admin, enables streamlined operations and better management of cases. This approach also fosters improved interdepartmental communication and faster response to new reports. Moreover, TrackP is built with ethical considerations at its core, addressing privacy and security issues associated with sensitive data. By restricting access to authorized personnel and ensuring accountability through system logs, TrackP supports law enforcement efforts responsibly and transparently.

To sum up, TrackP is a scalable & light-weight solution which might be helpful for law enforcement to locate missing people readily. The project still has a long way to go before things are ready for prime time, with many refinements and real-world tests ahead of it, but that potential could be an early step toward the next generation of tech-focused rescue ops. TrackP is not just a platform. it is one more step to make our society an integral part of technology (a new structure) which will contribute solving one of the most critical social problems: missing persons.

## REFERENCES


[1] Dr. J. Aruna Devi, Sagineti Lasya Priya, Vangalapudi Jeshitha, Sanjeevani Boddapati, Yendamuri Ganesh, Ratnala Mohan Krishna, “locating missing persons”, Journal of Nonlinear Analysis and Optimization, January 2024.

[2] Peace MuyamboAn “Investigation on the Use of LBPH Algorithm for Face Recognition to Find Missing People in Zimbabwe”, International Journal of Engineering Research & Technology (IJERT), July 2018.

[3] Mageswaran.S1 , Sivakumar.C2 , Prasanth.G3 , Mr. M. Sundaram ME.4, “Finding Missing Person Based on

Face Recognition Using AI in Video Surveillance System", International Journal for Research in Applied Science & Engineering Technology (IJRASET), May 2023.

[4] Divyansh Yadav1 , Janhvi Tyagi2 , Dipanshu3 , Ritu Tiwari4 , Ms. Pawan Pandey5, "Finding Missing Person/Child Using AI", International Journal of Research Publication and Reviews (IJRPR), May 2023.

[5] Pournami S. Chandran1, Byju N B2, Deepak R U3, Nishakumari K N4, Devanand P5 , Sasi P M6, "Missing Child Identification System using Deep Learning and Multiclass SVM", 2018 IEEE Recent Advances in Intelligent Computational Systems (RAICS), Dec 2018.

[6] Magnus D. Vigeland a, , Franco L. Marsico b , Mariana Herrera Pinero ˜ b , Thore Egeland c, "Prioritising family members for genotyping in missing person cases: A general approach combining the statistical power of exclusion and inclusion", Forensic Science International: Genetic, 2020.

[7] Mrs. Vishakha Shelke, Mr. Gunjan Mehta, Ms. Prajakta Gomase, "Searchious: Locating missing people using an optimised face recognition algorithm", IEEE Xplore, 2021.

[8] Fanny Coudert a,, Denis Butin b, Daniel Le Métayer c, "Body-worn cameras for police accountability: Opportunities and risks", ScienceDirect, 2015.

[9] Li Hou1,2,3, Wanggen Wan1,3, Jenq-Neng Hwang4, Rizwan Muhammad1,3, Mingyang Yang1,3 and Kang Han1,3, "Human tracking over camera networks: a review", EURASIP Journal, 2017.

[10] Muhammad Afnan Dzulkiflia, S.Sulaimana,b, A.Saparona,b, "Tracking System for Missing Person", IEEE Xplore, 2017.

[11] Mila Mileva and A. Mike Burton, "Face search in CCTV surveillance", Springer Open, 2019.

[12] Meduri Saketh1 · Neha Nandal2 · Rohit Tanwar3 · B. Pradyun Reddy4, "Intelligent surveillance support system", Discover Internet of Things, 2023.

[13] Viktoria R. Mileva , Peter J. B. Hancock and Stephen R. H. Langton, "Visual search performance in 'CCTV' and mobile phone-like video footage", Springer Open, 2021.

[14] Kristen A. Bakera, Sarah Laurencea,b, Catherine J. Mondlocha, "How does a newly encountered face become familiar? The effect of within-person variability on adults' and children's perception of identity", Cognition, 2017.

[15] Adelaide Bragias, Kelly Hine & Robert Fleet, "'Only in our best interest, right?' Public perceptions of police use of facial recognition technology", Routledge Taylor and Francis Group, 2021.

[16] Robert G. Alexander Gregory J. Zelinsky, "Visual similarity effects in categorical search", Journal of Vision, (2011).